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Evaluation Yourself

1. Calculate the ΔH^0_r for the reaction $CO_2(g) + H_2(g) \rightarrow CO(g) + H_2O(g)$

   Given that ΔH^0_f for $CO_2(g)$ and $H_2O(g)$ are -393.5, -111.31 and -242 kJ mol^{-1} respectively.

Solution :

Given

$\Delta H^0_f\, CO_2 = -393.5$ kJ mol^{-1}

$\Delta H^0_f\, CO = -111.31$ kJ mol^{-1}

$\Delta H^0_f\, (H_2O) = -242$ kJ mol^{-1}

$CO_2(g) + H_2(g) \rightarrow CO(g) + H_2O(g)$

$\Delta H^0_r = ?$

$\Delta H^0_r = \Sigma\, (\Delta H^0_f)_{products} - \Sigma\, (\Delta H^0_f)_{reactants}$

$\Delta H^0_r = [\Delta H^0_f\, (CO) + \Delta H^0_f\, (H_2O)] - [\Delta H^0_f\, (CO_2) + \Delta H^0_f\, (H_2)]$

$\Delta H^0_r = [-111.31 + (-242)] - [-393.5 + (0)]$

$\Delta H^0_r = [-353.31] + 393.5$

$\Delta H^0_r = 40.19$

$\Delta H^0_r = +40.19$ kJ mol^{-1}

2. Calculate the amount of heat necessary to raise 180g of water from to 25^0 C to 100^0 C. Molar heat capacity of water is 75.3 $Jmol^{-1}\, K^{-1}$.

Solution :

Given :

number of moles of water n

$= \frac{180\text{ g}}{18\text{ g mol}^{-1}} = 10$ mol

molar heat capacity of water

$C_P =$ 75.3 J K^{-1} mol^{-1}

$T_2 = 100^0$ C = 373 K

$T_1 = 25^0$ C = 298 K

$\Delta H = ?$

$\Delta H = nC_P\, (T_2 - T_1)$

$\Delta H = 10$ mol $\times$ 75.3 J mol^{-1} K^{-1} $\times$ (373 – 298) K

$\Delta H = 56475$ J

$\Delta H = 56.475$ kJ

3. From the following data at constant volume for combustion of benzene, calculate the heat of this reaction at constant pressure condition.

$$C_6H_6\,(l) + \frac{7}{2}O_2(g) \rightarrow 6\,CO_2\,(g) + 3\,H_2O\,(l)$$

ΔU at 25^0 C = –3268.12 kJ

Solution :

Given

$T = 25^0$ C = 298 K ;

$\Delta U = -3268.12$ kJ mol^{-1}

$\Delta H = ?$

$\Delta H = \Delta U + \Delta n_g RT$

$\Delta H = \Delta U + (n_p - n_r)\, RT$